

Pentasoftware Democratizes Secure and Scalable IoT Solutions Using AWS

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“AWS is, by far, the ideal enterprise cloud provider with scalability, reliability, security, and cost-effectiveness for our needs.”

—Jaime González, Chief Technology Officer

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Pentasoftware adopted 37 serverless AWS services to fulfill its goal of democratizing IoT solutions for SMBs. Pentasoftware is a cutting-edge software company specializing in affordable software-as-a-service solutions deployed exclusively on AWS. The company needed a scalable and agile cloud solution to develop its two keystone industrial monitoring solutions, Neuron and Akron. By using serverless AWS infrastructure like AWS IoT SiteWise and Amazon DynamoDB, Pentasoftware saves an average of 23 percent of costs for customers, serves API requests in under 1 second, maintains 99.999 percent availability, and streamlines onboarding for new customers.

For small and midsize businesses (SMBs), finding affordable Internet of Things (IoT) solutions can be a challenge, especially in the manufacturing and oil industries. Larger companies with economies of scale typically have little issue building an IoT infrastructure to monitor all their devices on the cloud in near real time. However, SMBs lack the same flexibility.

How can smaller companies level the playing field when costs are such a high barrier to entry? That's the problem [Pentasoftware](#) aims to solve with its suite of IoT solutions powered by Amazon Web Services (AWS).

Creating a Fully Serverless Business Model on AWS

Pentasoftware is a software company specializing in affordable software-as-a-service (SaaS) solutions for the industrial IoT field, created exclusively on AWS. It employs serverless technology, deployed on event-driven architecture and a collection of microservices. "We offer all our services in a pay-as-you-go scheme because affordability

is one of our main goals,” says Jaime González, chief technology officer at Pentasoft. “We want to be a key player in democratizing IoT onboarding to SMBs.”

Founded in 2008, Pentasoft originally offered on-premises desktop software, but it quickly expanded to take on new projects. One of those involved a large data warehouse that needed near-real-time advanced analytics, which motivated the company to transition to a SaaS business model. In 2014, the company shifted to a microservices architecture and began modernizing with serverless AWS solutions. In 2020, the Pentasoft team began building an IoT platform for its two flagship monitoring solutions, [Neuron](#) and [Akron](#), to support its SaaS operating model. Neuron provides powerful and affordable industrial monitoring on the cloud, and Akron provides monitoring and self-service solutions geared toward the oil and gas industry.

From 2008 to 2023, Pentasoft adopted 40 different serverless AWS services to build its solutions. By using AWS, Pentasoft can support a full SaaS business model and deliver software at scale. “AWS is, by far, the ideal enterprise cloud provider with scalability, reliability, security, and cost-effectiveness for our requirements,” says González. “It offers a complete set of advanced services and a vast selection for every need.”

Developing Near-Real-Time Monitoring Solutions Akron and Neuron Using AWS

Customers can seamlessly connect their physical sensors to Neuron, which securely ingests data and efficiently uploads it to the cloud. Pentasoft sets up this data ingestion using [AWS IoT Core](#), a service to simply and securely connect devices to the cloud. Neuron provides full telemetry and control of all industrial sensors, whether they measure pressure, temperature, activity, or other metrics. To model and store the data, Pentasoft uses [AWS IoT SiteWise](#), a managed service that simplifies modeling, collecting, and organizing industrial equipment data. “Agility is important because customers can’t afford to stop production to connect sensors or configure infrastructure,” says González. “We are now able to deploy or serve any API request in under 1 second.”

Neuron is a turnkey solution that works for virtually all sectors in the industry. Pentasoft continues to introduce new features, which users can [demo](#) through their web browser. For example, using Neuron, customers can create digital twins of their industrial environments. Pentasoft powers this feature using [AWS IoT TwinMaker](#), which provides the tools developers need to build digital twins to help optimize building operations, increase production output, and improve equipment performance.

Customer data is aggregated into dashboards, where users can extract insights through robust analytics. For this, Pentasoft integrates [Amazon QuickSight](#), which provides data-driven organizations with unified business intelligence at hyperscale. For setting up and deploying these solutions is a quick, simple process. It only

minutes for Pentasoft to onboard and provision new customer accounts. “Using serverless on AWS, everything grows and scales according to need,” says González. “The only thing to do is to help end users build up the dashboards and customize how they want to visualize their information.”

Akron focuses on the oil and gas industry and follows the same basic architecture as Neuron. However, instead of industrial sensors, Akron monitors electronic devices installed in petrol stations so that customers can monitor and control their fuel dispensers, fuel tanks, price boards, payment terminals, and many other devices. Getting data in near real time is crucial for customers in this industry. As all petrol stations in Spain upload their prices to a central repository, Pentasoft provides its customers with geolocated near-real-time competition prices, helping them simply optimize their pricing policy. To achieve this, Pentasoft uses [Amazon DynamoDB](#), a fully managed, serverless key-value NoSQL database designed to run high-performance applications at any scale. In doing so, Pentasoft can deliver 99.999 percent uptime of its solution. “Availability is a must,” says González. “Our customers need to know that they can use this application in its full functionality 24/7.”

Using serverless AWS solutions, Pentasoft is able to offer a pay-as-you-go pricing scheme for its SMB customers. As a result, the company has delivered customers an average of 23 percent cost savings.

Working Toward Making IoT Solutions Even More Affordable

Pentasoft continues to listen to its SMB customers and address their need for lower prices. In 2021, Pentasoft became an [AWS Partner](#). In doing so, the company received guidance from AWS on how to provide innovative solutions and deliver value to customers. Now, as an independent software vendor, Pentasoft offers Neuron through the AWS Marketplace. It plans to develop and offer more solutions there in the future. The company believes there is a path forward to more innovation to provide cost-effective IoT solutions to businesses of any size.

“We really believe in the fact that democratization means accessibility to new technology for all—not only for the big players, but for small companies and end users too,” says González.

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